

Automating Course Articulation: A Deep Metric Learning Framework Using Public Data

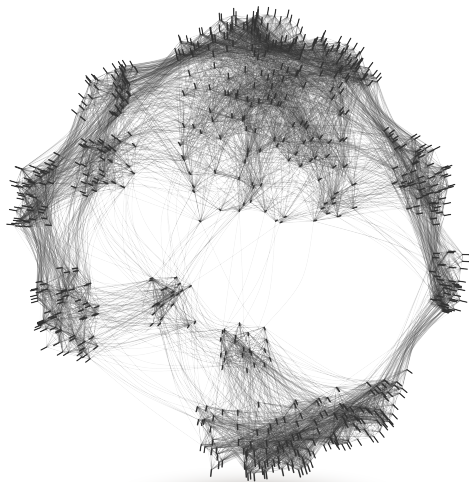
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The Problem: The Transfer Maze

- The process for determining course equivalency, or **articulation**, is a formidable, largely manual process that creates significant barriers for students [14].
- In California's public system alone, articulation officers at **149 individual campuses** manually negotiate and update agreements [4, 9, 7, 6].
- This task of “bleak combinatorics” is inefficient, slow, and inherently intractable, struggling to keep pace with the needs of a vast and mobile student body [14].
- This is not a niche issue; transferring between institutions has become a normative part of the modern student's academic journey [1].

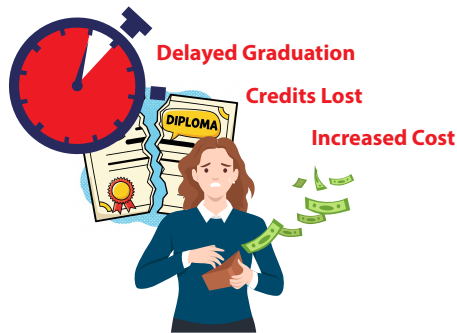


The High Cost to Students & Institutions

Consequences of an Inefficient System

The administrative friction of the transfer process creates a cascade of negative consequences that fall almost entirely on students.

- **Significant Credit Loss:** On average, transfer students lose an estimated **43%** of their academic credits [21, 1].
- **Increased Time-to-Degree:** Lost credits directly delay graduation and postpone entry into the workforce [21].
- **Greater Financial Burden:** Repeating courses increases tuition costs and can exhaust a student's financial aid eligibility [21, 5].
- **Reduced Student Persistence:** The frustration of the process contributes to lower graduation rates for transfer students compared to their non-transfer peers [16].

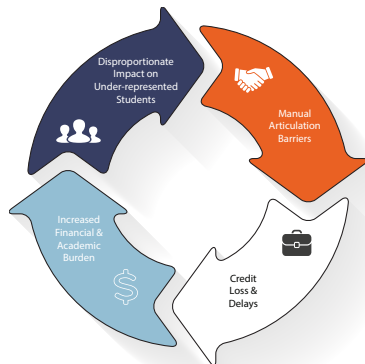


A Critical Equity Issue

This is not just an administrative problem; it's an equity problem.

The barriers imposed by an inefficient articulation system fall most heavily on the very students institutions are striving to support [20].

- Low-income and underrepresented students disproportionately rely on transfer pathways from community colleges [20].
- Recent transfer enrollment growth has been driven primarily by Black and Hispanic students [8].
- This creates a **feedback loop**: transfer barriers cause credit loss, imposing burdens that undermine efforts to close equity gaps [20, 8].
- Therefore, automating articulation is not just an operational optimization; it is a **necessary intervention** to foster educational equity [5].

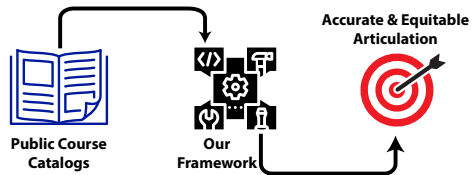


The Goal & Our Contribution

The Goal

To develop and validate a novel framework that automates course articulation using only publicly available data.

The resulting system must be:



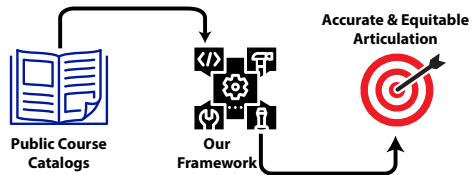
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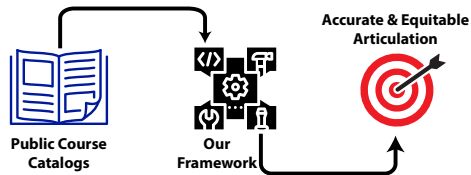
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- 1 **A Highly Accurate Framework:**
Developed a complete pipeline achieving state-of-the-art accuracy on real-world data.
- 2 **An Innovative Feature Vector:** Designed a novel composite vector combining local and global semantics to improve classification.



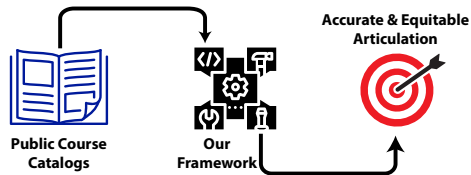
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The resulting system must be:

- Accurate & Scalable
- Computationally Efficient
- Inherently Privacy-Preserving



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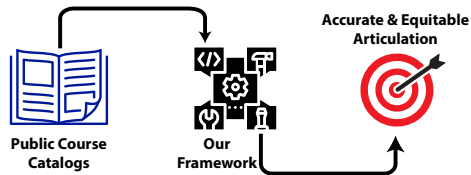
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Developed a complete pipeline achieving state-of-the-art accuracy on real-world data.
- 2 **An Innovative Feature Vector:** Designed a novel composite vector combining local and global semantics to improve classification.
- 3 **An Efficient & Private Approach:**
Created a decoupled solution that avoids the high costs and privacy concerns of prior methods.

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The Landscape of Automation

Prior attempts at automation have evolved, with each generation introducing new capabilities while also exposing new limitations.

Approach	Key Characteristic	Core Limitation
Keyword & Statistical (TF-IDF)	Weight terms based on statistical importance [2].	No semantic understanding; cannot grasp synonyms or context.
Static Embeddings (word2vec, GloVe)	Represent words as averaged, pre-trained vectors.	Context-insensitive, and averaging vectors loses critical semantic information.
Enrollment-Based (course2vec)	Learn similarity from student co-enrollment patterns [15].	Requires sensitive student data, raising major privacy and generalizability issues [19].
Direct LLM Classification	Use a large language model as an end-to-end classifier.	High computational cost, opaque "black box" reasoning, and sensitive to prompt phrasing [11]. Also empirically illustrated in our preliminary work (Phase I)

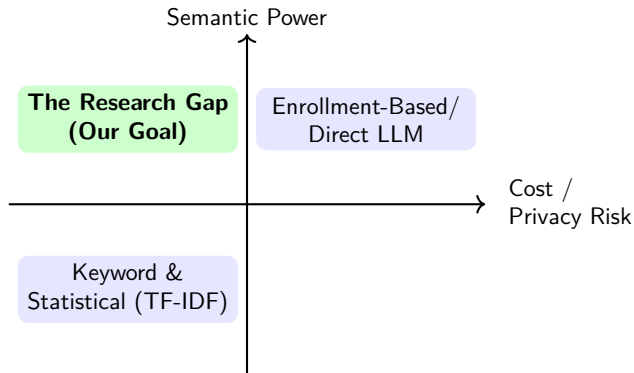
The Research Gap

A review of prior work reveals a fundamental trade-off: as models gain semantic power, they tend to become more computationally intensive, less interpretable, or more demanding of specialized or private data.

The Opportunity

The limitations of direct LLM classification (cost, opacity) and enrollment-based methods (privacy, limited access) point toward a gap in the existing research for a new paradigm [14, 19].

An effective solution must harness the semantic power of large models without inheriting their operational burdens.

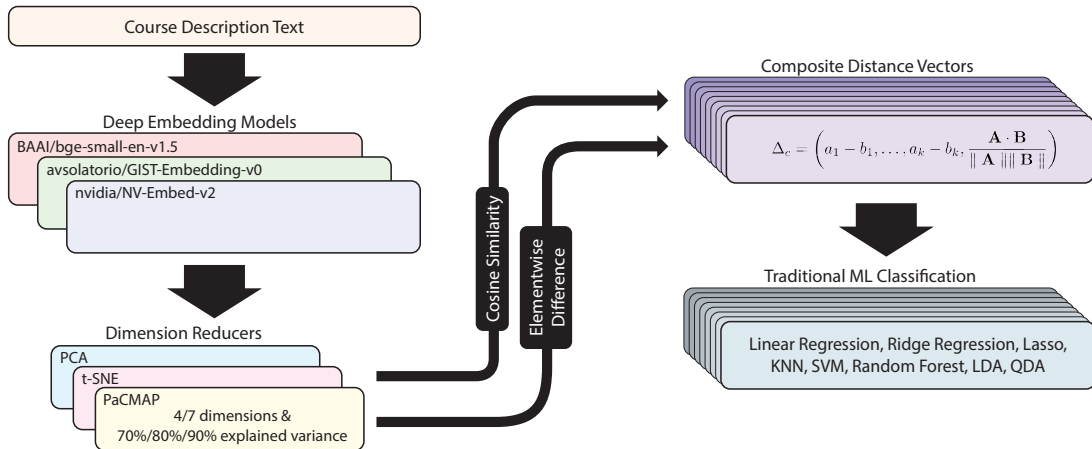


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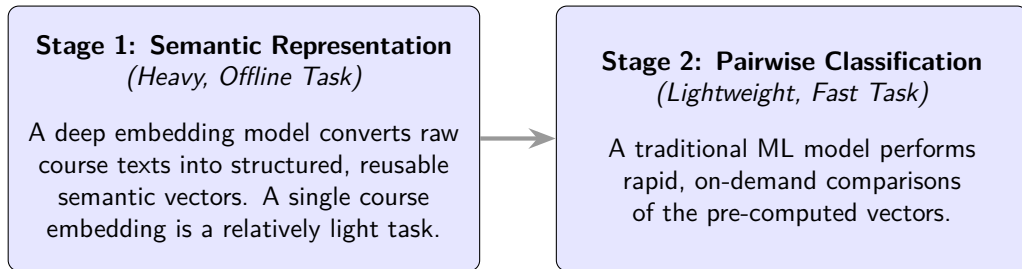
A Decoupled Framework: High-Level Architecture

Our framework's core principle is to **decouple rich semantic representation from the final classification task**. This creates a more efficient, scalable, and transparent system.



Core Principle: Decoupling Representation from Classification

By separating the process into two stages, we gain the semantic power of deep learning while avoiding the high operational costs of end-to-end LLM classification [11] and the privacy risks of enrollment-based methods [19].



Prompting Methods: From Verbose to Normalized

The distinction between two approaches

Phase 1: Direct LLM Classification

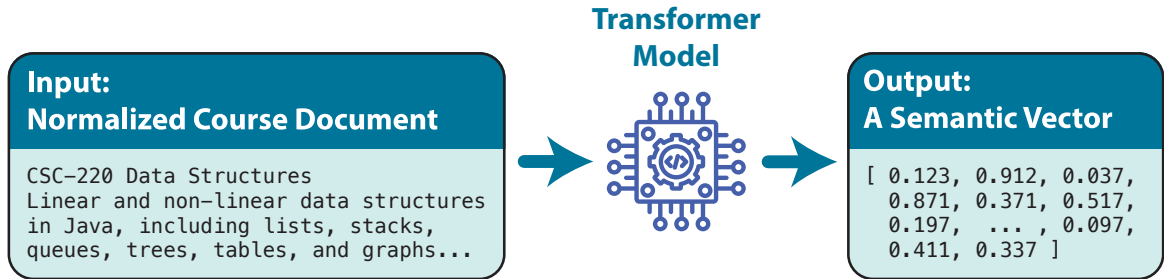
- **Method:** End-to-end classification using a large language model.
- **Input:** A single, lengthy prompt containing verbose instructions, few-shot examples, and the raw course descriptions.
- **Challenge:** Performance was highly sensitive to prompt phrasing and formatting, aligning with known limitations of LLMs. This method was not robust for a production system.

Phase 2: Decoupled Framework

- **Method:** Generate embeddings from text, then classify.
- **Input:** Normalized course text (e.g., “CSC-220 Data Structures. . .”), with no external instructions.
- **Benefit:** The model learns semantic meaning directly from the text itself, making it robust to formatting changes and eliminating the need for complex prompt engineering. This is more suitable for a scalable system.

Step 1: Deep Contextual Embeddings

The first step is to convert unstructured course catalog text into a structured, semantically rich vector using a pre-trained transformer model [10, 17].



Step 2: Our Novel Feature Vector (Δ_c)

To classify course pairs, we need features that represent the *relationship* between them. We designed a novel **composite distance vector** (Δ_c) to provide the classifier with a richer, more discriminative feature set.

Combining Local & Global Information

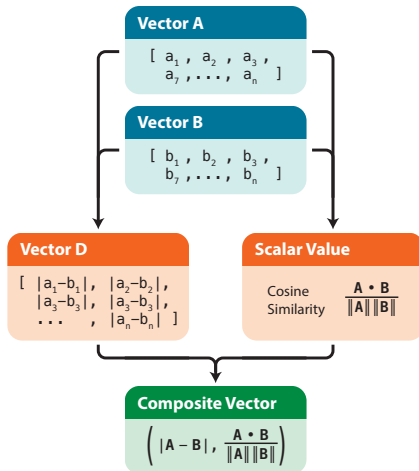
The vector combines two distinct types of information:

- **Local Disparities:** The granular, element-wise difference between the two course vectors.
- **Global Alignment:** A single, holistic score of their overall similarity in the semantic space.

The Formula

For two k -dimensional course vectors, A and B , the composite vector Δ_c is constructed by concatenating their element-wise difference with their cosine similarity:

$$\Delta_c = \left(|a_1 - b_1|, \dots, |a_k - b_k|, \frac{A \cdot B}{\|A\| \|B\|} \right)$$



Step 3: Domain-Specific Fine-Tuning

General-purpose models lack the specialized “vocabulary” for academic text. To create a more discriminative embedding space, we fine-tune a pre-trained model on our course data using **deep metric learning**.

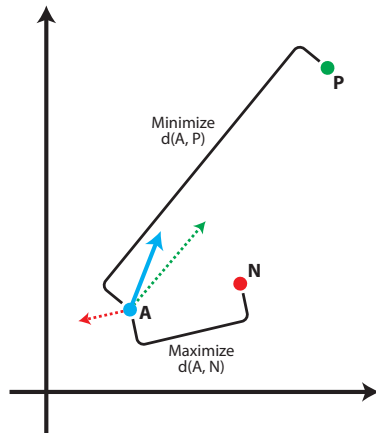
Learning Objective: The Triplet Loss

We train the model using a **Triplet Loss** function, which teaches the model to understand nuanced similarity by operating on triplets of courses [18, 13]:

- An **Anchor** course (A)
- A **Positive**, equivalent course (P)
- A **Negative**, non-equivalent course (N)

The goal is to adjust the embedding space such that the distance between the Anchor and Positive is smaller than the distance between the Anchor and Negative, enforced by a margin (α):

$$L(A, P, N) = \max(d(A, P) - d(A, N) + \alpha, 0)$$



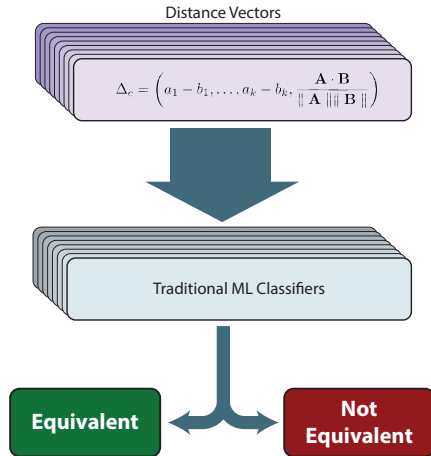
Step 4: Downstream Classification

The final step is to feed the engineered composite distance vectors (Δ_c) into a traditional machine learning model to produce the final equivalency prediction.

Systematic Model Evaluation

To identify the most effective algorithm for this task, we systematically evaluated a comprehensive suite of models, including representatives from major algorithmic families:

- Linear Models (e.g., Logistic Regression)
- Kernel-Based Models (e.g., SVM)
- Instance-Based Models (e.g., KNN)
- Ensemble Tree Models (e.g., Random Forest)
- Gradient Boosting Models (e.g., XGBoost)

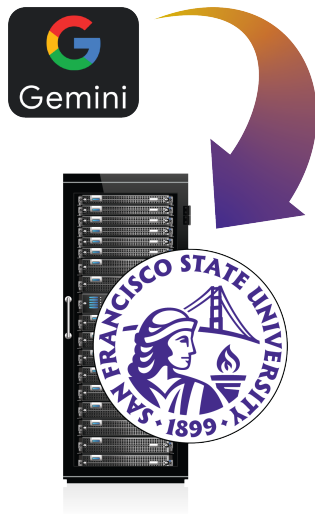


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The Experimental Environment

- **Hybrid Computing Approach:** Utilized both a cloud-based LLM and a local High-Performance Computing (HPC) cluster.
- **Cloud for Baseline:** Google's Gemini v1.0 was used for the initial, exploratory direct LLM classification.
- **Local HPC for Core Work:** The SFSU POLARIS HPC cluster was used for all primary, computationally intensive experiments.
 - **Embedding & Fine-Tuning:** GPU tasks on 4x NVIDIA A100s
 - **Classifiers:** CPU-bound tasks on AMD EPYC 9534 CPUs
- **Software:** The pipeline was built with Python, PyTorch, Hugging Face, scikit-learn, and XGBoost.



Datasets & Evaluation

The framework was trained and validated on a real-world dataset to ensure the results are robust and generalizable.

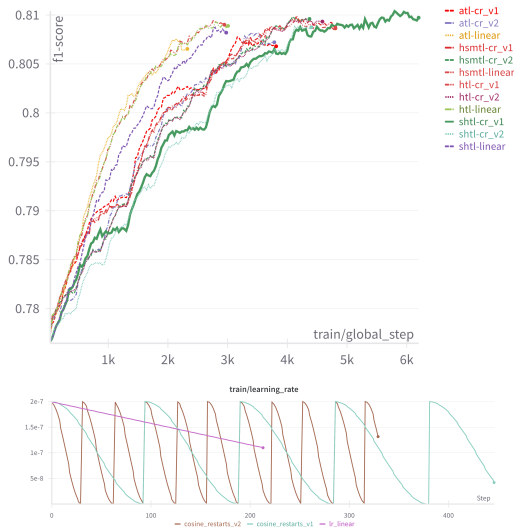
Characteristic	Initial Dataset	PPM Corpus
Source	Manually curated via ASSIST	Program Pathways Mapper (PPM)
Purpose	Preliminary screening, prototyping, and initial classifier evaluation	Definitive fine-tuning and final pipeline evaluation
Ground Truth	Established articulation agreements	Course Identification Number (C-ID)
Final Size	400 course pairs	9192 course pairs (test set)
Partitioning	Stratified random sample	Stratified 50/50 train/test split

Adapting a Model with Domain-Specific Fine-Tuning

The Process

We adapted a general-purpose model by fine-tuning it on the PPM Corpus using deep metric learning.

- **Objective:** Batch Triplet Loss functions were used to teach the model the nuanced semantics of academic text.
- **Optimization:** We paired a stable AdamW optimizer with a Cosine Annealing learning rate schedule to effectively navigate the complex loss landscape.



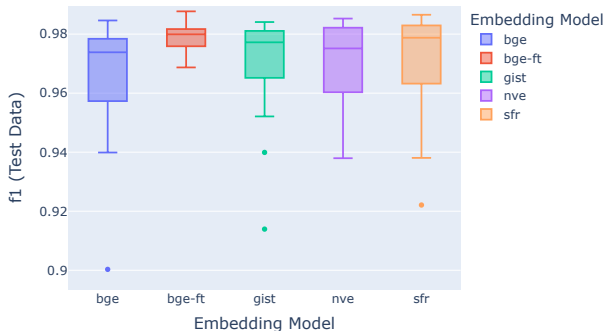
Finding 1: Fine-Tuned Model is Statistically Superior

The result of the fine-tuning process is a model that is demonstrably more accurate and consistent on the held-out test data.

Key Results

- Our fine-tuned model (**bge-ft**) had the highest mean F_1 -score ($\mu = 0.9786$) and the lowest variance.
- A one-way ANOVA and Games-Howell post-hoc test confirmed that the performance gap is statistically significant against *all* other models.
- This includes models that were orders of magnitude larger.

Mean f1-Score Distribution by Embedding Model (Test Data)

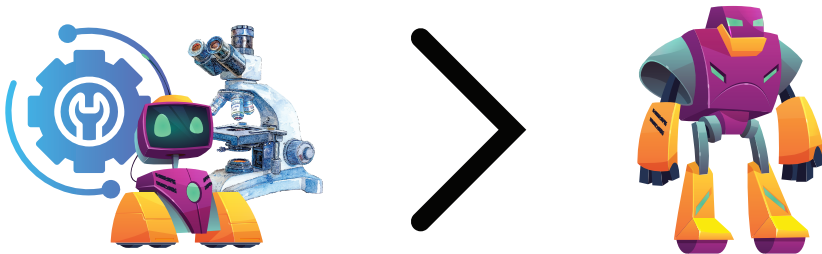


Key Insight: Adaptation Outperforms Scale

But “So What?”

For specialized domains like academic text, creating a bespoke embedding space through targeted fine-tuning is more effective than relying on sheer model scale.

The fine-tuning process retrained the model’s attention mechanism, teaching it the specific semantics required to make fine-grained distinctions that larger, general-purpose models may miss.



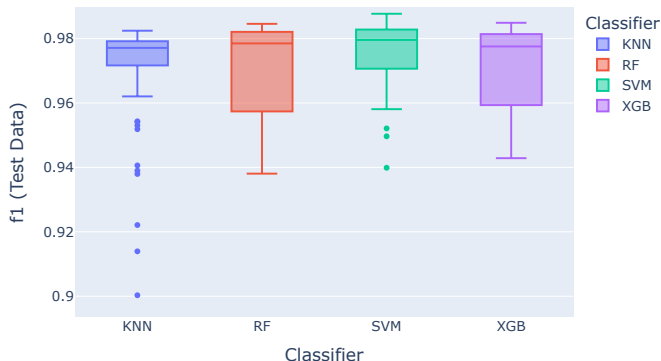
Finding 2: All Finalists Perform Exceptionally Well

The first key result from the classifier evaluation is that our feature engineering and fine-tuning were highly effective, leading to exceptional performance across all finalist models.

High & Stable Performance

- All four finalist classifiers (SVM, RF, XGBoost, KNN) achieved high and stable F1-scores.
- As the boxplot shows, the distributions are tightly clustered with mean scores approaching or exceeding **0.97**.
- This demonstrates the robustness of the upstream feature representation.

Mean f1-Score Distribution by Classifier (Test Data)



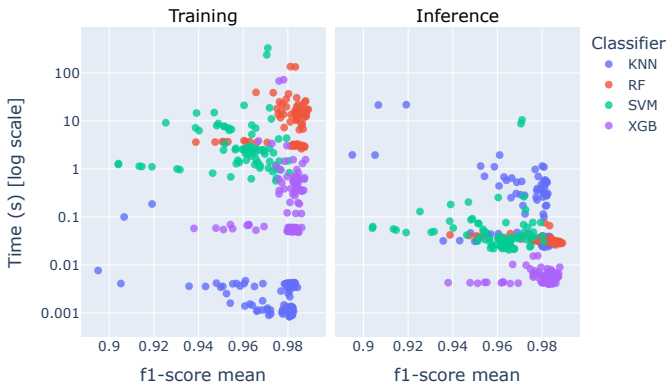
The Trade-Off: Peak Accuracy vs. Operational Efficiency

While all models performed well, a deeper analysis reveals a classic trade-off, leading to context-dependent recommendations for deployment.

Key Finding

- **For Optimal Efficiency:** **Random Forest (RF)** and **XGBoost** were nearly as accurate but an order of magnitude faster and more predictable at inference time.
- **For Maximum Accuracy:** The **Support Vector Machine (SVM)** was the statistical winner, proving to be the most accurate and consistent classifier.

Comparing Model f1-Score vs. Time Costs



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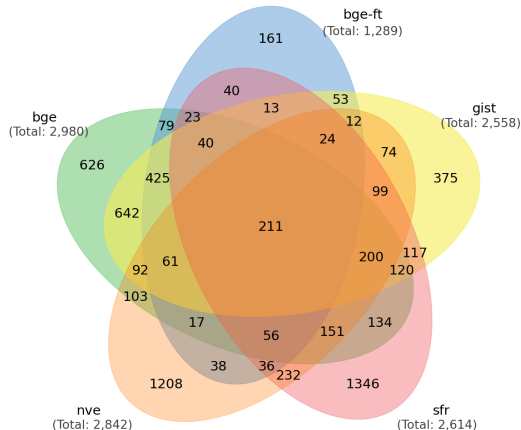
Qualitative Analysis: Why Do Errors Still Occur?

High-level metrics don't tell the full story; a purely numerical analysis can be misleading [12].

Finding: The Bottleneck is Data, Not the Model

Our analysis revealed that most errors are not random, but are systematic issues rooted in the source data itself.

- A core set of **211 "hard" pairs** were misclassified by *every single model* we evaluated.
- This proves the primary bottleneck for performance has shifted from being model-centric to **data-centric**.



Shared Misclassifications: A Data-Centric Problem

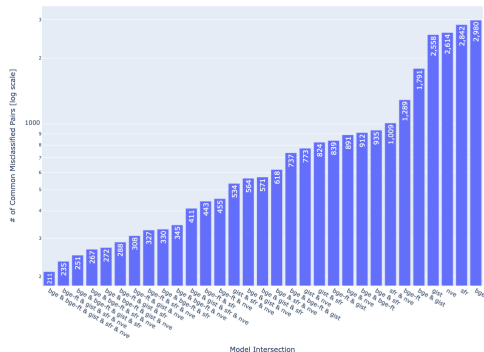
To diagnose the source of errors, we analyzed their overlap across all models. The results provide strong evidence that the errors are systematic.

Failures are Systematic, Not Random

The analysis of misclassifications reveals a high degree of overlap across all evaluated models.

- A significant portion of failures are systematic products of the source data itself.
- These "hard" examples consistently challenge a wide range of semantic models, from small specialists to large generalists.
- This indicates errors stem from inherent data challenges—like ambiguity or annotation artifacts—not model weaknesses.

Overlap of Misclassified Pairs Between Embedding Models



Root Cause Analysis: False Negatives (Missed Equivalencies)

A False Negative occurs when the system fails to identify a true, existing equivalence. This represents a missed opportunity for a student.

Causes

- **Semantic Divergence:** Officially equivalent courses are described with vastly different terminology or pedagogical focus. The model correctly sees the texts as dissimilar; the error is in the inconsistent source data.
- **Minimalist Descriptions:** One or both course descriptions are too sparse or incomplete to provide enough textual signal for the model to find a confident match.

Example: Semantic Divergence

Course A "... examines... developmental milestones from middle childhood through adolescence... "

Course B "... examines... diversity and inclusion... anti-bias curriculum... promote inclusive... classroom... "

Result: False Negative
(Model correctly sees texts as different)

Root Cause Analysis: False Positives (Incorrect Matches)

A False Positive occurs when the system incorrectly classifies two non-equivalent courses as equivalent. This is a harmful error that could mislead a student.

Causes

- **Topical Overlap:** Courses cover the same broad subject but differ critically in academic level or their position in a sequence (e.g., Physics I vs. Physics II). The model correctly identifies high topical similarity but can't infer the sequence.
- **Vague Descriptions:** Descriptions use generic language, lacking the specific detail needed for differentiation. This is a known challenge in short-text semantic similarity [3].

Example: Topical Overlap

- | | |
|-----------------|--|
| Course A | PHYS-4A: <i>"...a systematic introduction to the principles of classical mechanics..."</i> |
| Course B | PHYS-4D: <i>"...a systematic introduction to the principles of modern physics..."</i> |

Result: False Positive
(Model sees high topical overlap)

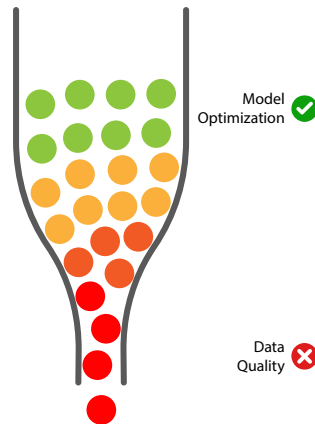
Conclusion of Analysis: The Primary Bottleneck

The qualitative analysis leads to a critical insight regarding the future of automated articulation.

The Bottleneck has Shifted from Model-Centric to Data-Centric

With an optimized pipeline, the limiting factor may no longer be the model's architecture or semantic capability.

- The model is performing correctly; it accurately reports when two texts are not semantically similar.
- The remaining errors are artifacts of the source data itself: inconsistent descriptions, vague language, and information gaps.
- Therefore, the most promising path to further improvement lies not in novel architectures, but in methodologies that directly address the quality and consistency of the input data [12].



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Limitations

While the proposed framework represents a significant advance, it is essential to acknowledge the boundaries of the current study.

Key Limitations

Performance is Capped by Data Quality:

The system's performance is fundamentally limited by the quality and content of the public course descriptions. It cannot infer information that is absent from vague, minimalist, or inconsistent source texts.

Generalizability of the Fine-Tuned Model:

The specialized *bge-ft* model was tuned on data from California's public colleges. Its performance may not be as high "out-of-the-box" in other contexts (e.g., private or non-US institutions) without re-tuning on local data.

Handling of Complex Articulation Rules:

The framework simplifies articulation into a binary classification of course pairs and does not natively handle complex one-to-many or many-to-many agreements, a challenge that persists for many automated systems [14].

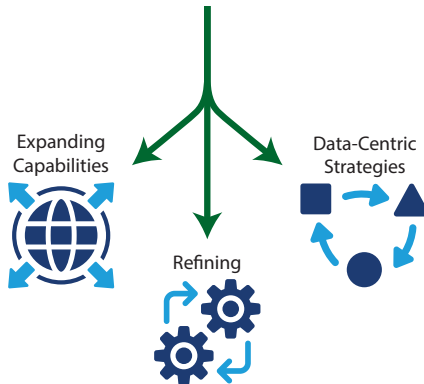
Future Work

The findings and limitations of this study give rise to several promising avenues for future research.

Key Research Directions

- **Data-Centric AI Strategies:** Focus on data quality via human-in-the-loop systems and dynamic data augmentation.
- **Expanding Capabilities:** Evolve the framework into a full course recommendation engine and use graph-based methods for complex articulation rules.
- **Refinement of Current Work:** Refine the core ML pipeline by exploring new feature combinations, multi-modal learning, and task-optimized loss functions.

Future Directions



Summary of Contributions

This research confronted the challenge of manual course articulation by designing, developing, and validating a novel computational framework.

Primary Contributions

- 1 **A Novel, Accurate, and Scalable Framework:** We developed an end-to-end pipeline that successfully automates course articulation using only public data, achieving state-of-the-art accuracy.
- 2 **Proof that Adaptation Outperforms Scale:** We proved that for this specialized domain, fine-tuning a smaller model for semantic nuance is statistically superior to relying on sheer model scale.
- 3 **A Practical Tool for Educational Equity:** We delivered a practical, computationally efficient, and privacy-preserving tool that can reduce administrative burden and help mitigate the systemic inequities faced by transfer students.

Thank You!!!

Questions?

Contact Information

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